§ 96. Discriminants of the orders.

1. *Definition.* If Q is a natural number, then the totality of the integral numbers of the quadratic field Ω which are congruent, modulo Q, to a rational number is called an *order*, and Q is called the conductor of this order. The order with conductor Q is denoted by $[Q]$.

From this definition it follows:

1. All rational integers belong to every order;
2. The sum, difference, and product of two numbers of the order belong to the same order;

that is, within the order addition, subtraction, and multiplication can be performed without restriction, but division cannot in general be so performed.

The totality of all integral numbers of the field Ω likewise forms an order (of conductor 1), the principal order, which would therefore be denoted by [1]. [1]

If Δ is the fundamental number of the field Ω, we put, according as Δ is even or odd,

$$\theta_0 = -\frac{1}{2}\sqrt{\Delta}, \qquad \text{or} \qquad \theta_0 = \frac{-1+\sqrt{\Delta}}{2}. \tag{1}$$

Then, by § 90, every integral number of the field Ω is representable in the form

$$\omega = x + y\theta_0, \tag{2}$$

where x and y are integers; and this number is congruent, modulo Q, to a rational number a if and only if y is divisible by Q, since then

$$\frac{x - a + y\theta_0}{Q}$$

must be an integer. Accordingly all numbers of the order $[Q]$ are contained in the form

$$\omega = x + yQ\theta_0,$$

or, in the two cases distinguished in (1), if

$$D = Q^2\Delta \tag{3}$$

is put:

$$\begin{aligned} & x - y\frac{\sqrt{D}}{2}, && \Delta \text{ even}, \\ & x - \frac{Q}{2}y + y\frac{\sqrt{D}}{2}, && \Delta \text{ odd}, \ D \text{ even}, \\ & x - \frac{Q-1}{2}y + y\frac{-1+\sqrt{D}}{2}, && D \text{ odd}. \end{aligned} \tag{4}$$

Thus, if we put

$$\theta = \frac{\sqrt{D}}{2}, \quad D \equiv 0 \pmod 4, \qquad \theta = \frac{-1+\sqrt{D}}{2}, \quad D \equiv 1 \pmod 4, \tag{5}$$

and replace in the two last formulae (4) the arbitrarily integral numbers

$$x - \frac{Q}{2}y, \qquad x - \frac{Q-1}{2}y$$

[1] Dedekind introduced the expression *order* in the general theory of algebraic numbers because, in the special case of the quadratic field, these orders correspond to Gauss's orders of quadratic forms. Hilbert uses for this the expression "ring" or "number ring". Cf. Dirichlet–Dedekind, *Vorlesungen über Zahlentheorie* (§ 172 of the third, § 170 of the fourth edition). Dedekind: *Über die Anzahl der Idealklassen in den verschiedenen Ordnungen eines endlichen Körpers*, Festschrift zur Säkularfeier von Gauss' Geburtstag, Braunschweig, 1877. Hilbert: *Die Theorie der algebraischen Zahlen*, Bericht der Deutschen Mathematischen Vereinigung, 1894/95, Chap. IX.

by x, we obtain the result:

2. *Theorem.* Every number of the order $[Q]$, and only these, is contained in the form

$$\omega = x + y\theta, \tag{6}$$

where x, y are arbitrary rational integers.

We call the system

$$(1, \theta) \tag{7}$$

a basis of the order $[Q]$.

By substituting (5), the numbers (6) may also be represented in the form

$$\omega = \frac{x + y\sqrt{D}}{2}, \tag{8}$$

where x, y are integers satisfying the condition

$$x^2 - y^2 D \equiv 0 \pmod 4. \tag{9}$$

The number D is a discriminant whose stem is Δ. We call it the *discriminant of the order* $[Q]$.